Lab 7: Fuzzing for Input Validation Bugs

BHUVAN KUMAR P · PES1UG24AM069 · Assigned repo: SETAPESU26/L6_L7_07_hospital_traige_priority (Lab 7) · Tools: Python 3.11, pytest 9.2.0, Hypothesis 6.170.15

The program `processor.py` parses three kinds of text input: patient vitals (`HR,SpO2,Temp`), BMI (`weight/height`) and ward-bed codes (`ICU-05`). `test_processor.py` was completed with Hypothesis property-based tests. Each function is fed random text, `None` and almost-valid inputs, and must never raise an exception. Three more tests check that valid input still gives the correct result.

Run	Result
Fuzz tests vs buggy `processor.py`	**6 failed**, 3 passed (all 6 crash tests failed)
Fuzz tests vs fixed `processor.py`	**9 passed** (also 9 passed at 5,000 examples per test)

Step 2: Run the buggy application

```
python processor.py
```

With valid input the buggy program works. With the 2-value input `72,98` it crashes with an `IndexError`.

Terminal - Step 2: python processor.py (buggy)

```
~/SE-Lab/Lab-7$ python processor.py
==== Running parse_vitals_summary ====
Enter vitals (HR,SpO2,Temp): 72,98,36.8
Summary: Vitals: HR=72.0, SpO2=98.0% - STABLE

==== Running compute_patient_bmi ====
Enter weight/height (e.g. 70/1.75): 70/1.75
BMI: 22.86

==== Running assign_ward_bed ====
Enter allocation (e.g. ICU-05): ICU-05
Assignment: Intensive Care Unit Bed #5
~/SE-Lab/Lab-7$ python processor.py
==== Running parse_vitals_summary ====
Enter vitals (HR,SpO2,Temp): 72,98
Traceback (most recent call last):
  File "/home/user/se-lab/Lab-7/processor.py", line 41, in <module>
    main()
  File "/home/user/se-lab/Lab-7/processor.py", line 30, in main
    print("Summary:", parse_vitals_summary(v_input))
                      ^^^^^^^^^^^^^^^^^^^^^^^^^^^^^
  File "/home/user/se-lab/Lab-7/processor.py", line 5, in parse_vitals_summary
    temp = float(parts[2])
                 ~~~~~^^^
IndexError: list index out of range
```

Step 4: Run the completed fuzz tests on the buggy code (failing)

```
python -m pytest test_processor.py
```

The full output (424 lines of Hypothesis tracebacks) was saved to `fuzz_results_buggy.txt`. The commands below show its start, every crashing input Hypothesis found, and the summary: 6 failed, 3 passed.

Terminal - Step 4: python -m pytest test_processor.py (buggy processor)

```
~/SE-Lab/Lab-7$ python -m pytest test_processor.py > fuzz_results_buggy.txt
~/SE-Lab/Lab-7$ head -n 7 fuzz_results_buggy.txt
============================= test session starts ==============================
platform linux -- Python 3.11.18, pytest-9.2.0, pluggy-1.7.0
rootdir: /home/user/se-lab/Lab-7
plugins: hypothesis-6.170.15
collected 9 items

test_processor.py FFFFFF...                                              [100%]
~/SE-Lab/Lab-7$ grep -o 'crashed on [^{].*' fuzz_results_buggy.txt | sort -u
crashed on '' with: ValueError("could not convert string to float: ''")
crashed on '' with: ValueError('not enough values to unpack (expected 2, got 1)')
crashed on '-' with: ValueError("invalid literal for int() with base 10: ''")
crashed on '/' with: ValueError("could not convert string to float: ''")
crashed on '/0' with: ValueError("could not convert string to float: ''")
crashed on '0' with: IndexError('list index out of range')
crashed on '0/' with: ValueError("could not convert string to float: ''")
crashed on '0/0' with: ZeroDivisionError('float division by zero')
crashed on 'ER' with: ValueError('not enough values to unpack (expected 2, got 1)')
crashed on None with: AttributeError("'NoneType' object has no attribute 'split'")
crashed on None with: AttributeError("'NoneType' object has no attribute 'strip'")
~/SE-Lab/Lab-7$ tail -n 8 fuzz_results_buggy.txt
=========================== short test summary info ============================
FAILED test_processor.py::test_parse_vitals_summary_no_crash - ExceptionGroup...
FAILED test_processor.py::test_compute_patient_bmi_safe - ExceptionGroup: Hyp...
FAILED test_processor.py::test_assign_ward_bed_safe - ExceptionGroup: Hypothe...
FAILED test_processor.py::test_parse_vitals_structured_no_crash - AssertionEr...
FAILED test_processor.py::test_compute_patient_bmi_structured_no_crash - Exce...
FAILED test_processor.py::test_assign_ward_bed_structured_no_crash - Assertio...
======================== 6 failed, 3 passed in 1.04s ========================
```

Bugs found by fuzzing and how they were fixed

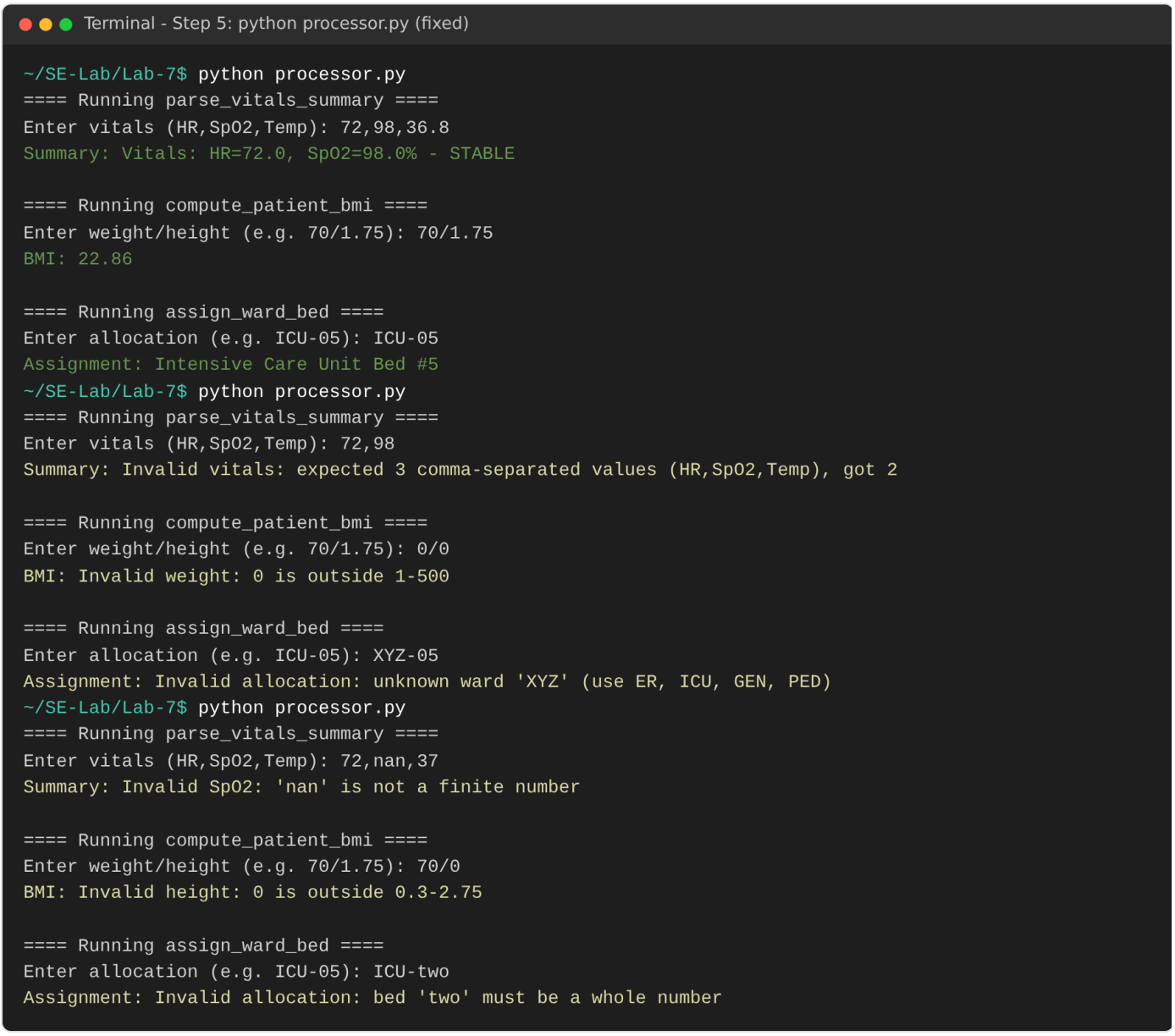
Function	Example input	Failure	Fix
All three	`None`	AttributeError on `.split()` / `.strip()`	Reject anything that is not a string
parse_vitals_summary	`'0'`, `'0,0'`	IndexError: list index out of range	Require exactly 3 comma-separated values
parse_vitals_summary	`''`, `'0,0,'`	ValueError: could not convert string to float	Parse each value safely; report which one is bad
parse_vitals_summary	`'nan,nan,nan'` (no crash)	Silently reported as STABLE	Reject non-finite values and values outside plausible ranges
compute_patient_bmi	`''`, `'70'`	ValueError: not enough values to unpack	Require exactly one '/'
compute_patient_bmi	`'/'`, `'0/'`, `'/0'`	ValueError: could not convert string to float	Parse weight and height safely
compute_patient_bmi	`'0/0'`, `'70/0'`	ZeroDivisionError: float division by zero	Weight 1–500 kg, height 0.3–2.75 m (finite)
assign_ward_bed	`''`, `'ER'`	ValueError: not enough values to unpack	Require exactly one '-'
assign_ward_bed	`'-'`, `'ER-0.0'`	ValueError: invalid literal for int()	Bed must be ASCII digits, 1–9999
assign_ward_bed	`'XYZ-05'`	KeyError on the ward lookup	Check the ward against the known list

Every function now returns a message starting with `Invalid` for bad input instead of raising, and gives the same result as before for valid input.

Step 5: Run the fixed application

`python processor.py`

Valid input gives the same results as before. The inputs that crashed it, plus `0/0`, an unknown ward, `nan`, a zero height and a non-numeric bed, now get a clear message instead of a traceback.



Step 6: Re-run the fuzz tests on the fixed code (passing)

`python -m pytest test_processor.py`

All 9 Hypothesis tests pass against the fixed `processor.py`.

